Loan Default Prediction System

A Comparative Study of Classification Algorithms

Mini Project
Supervised Learning Module

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Dataset: Module 3 Project Data.csv

ABSTRACT

Banks provide loans to customers every day, but there is always a possibility that some customers may not be able to repay them. Identifying such customers in advance can help banks reduce financial losses and make better lending decisions.

The main aim of this project is to develop a Loan Default Prediction System using machine learning. The system uses customer information, including employment status, bank balance, and annual salary, to predict whether a customer is likely to default on a loan.

For this project, four machine learning algorithms are used: Logistic Regression, Random Forest, Support Vector Machine (SVM), and XGBoost. Their performance is compared using accuracy, precision, recall, F1-score, and ROC-AUC.

After comparing the results, Logistic Regression is selected as the best model because it achieves the highest ROC-AUC score of approximately 0.9487. Feature importance analysis also shows that bank balance plays the most significant role in the predictions made by the tree-based models.

1 Problem Statement

This project addresses the problem by using machine learning to predict loan default based on a customer's employment status, bank balance, and annual salary.

The problem is treated as a binary classification task:

- **0 — No Default:** The customer did not default on the loan.
- **1 — Default:** The customer defaulted on the loan.

The model learns patterns from existing customer records and uses them to predict the outcome for new customer data.

Key Idea

An important challenge in this project is that most customers in the dataset did not default, while only a small number did. Because of this imbalance, a model could achieve high accuracy simply by predicting that almost everyone will repay their loan.

2 Dataset Description

The project uses a banking dataset containing 10,000 customer records. The dataset "Module 3 Project Data.csv" is loaded into Python using the Pandas library and examined before training the machine learning models.
The dataset contains the following columns:

Column	Description
Index	Record identifier, which is removed before model training
Employed	Indicates whether the customer is employed (1) or not (0)
Bank Balance	Current bank balance of the customer
Annual Salary	Annual income of the customer
Defaulted?	Indicates whether the customer defaulted (1) or did not default (0)

For prediction, three columns are used as input features: Employed, Bank Balance, and Annual Salary. The Defaulted? column is used as the target variable.

Before training, the dataset is checked for missing values, incorrect data types, duplicate records, and other data-quality issues. The Index column is removed because it does not provide useful information for predicting loan default.

2.1 Class Balance

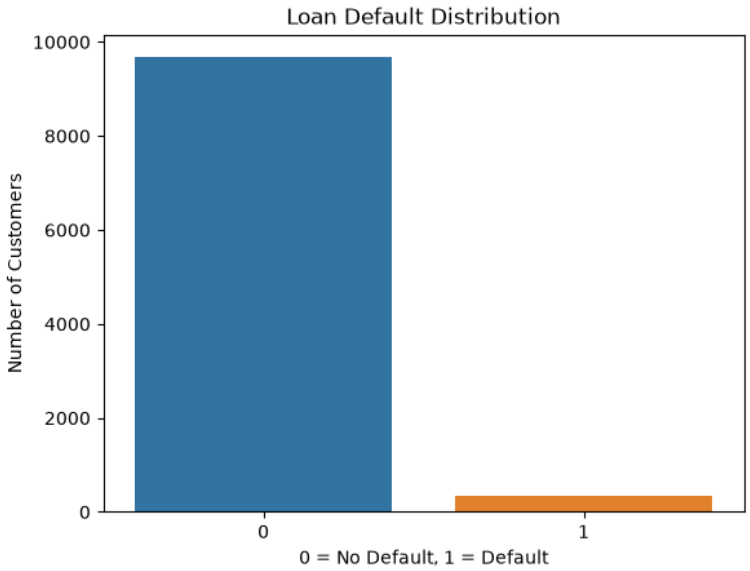


Figure 1: Class distribution of the target variable. Defaults make up only a small fraction of the dataset.

2.2 Feature Correlations

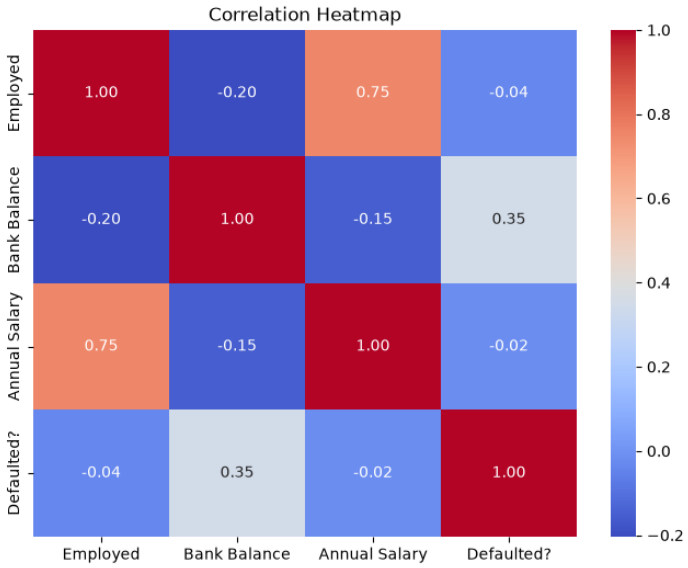


Figure 2: Correlation heatmap of the numeric features and the target.

3 Methodology

3.1 Data Preprocessing

Data preprocessing is an important step because the quality of the input data can affect the performance of machine learning models.

The following steps are performed in this project:

1. The dataset is loaded using Pandas.
2. The Index column is removed because it is not useful for predicting loan default.
3. The input features and target variable are converted into numerical format.
4. Infinite values are replaced with missing values.
5. Missing values in the input features are filled using the median of each column.
6. Rows with missing target values or target values other than 0 and 1 are removed.
7. The cleaned data is prepared for model training.

3.2 Handling Class Imbalance

Since only 3.33% of the customers defaulted, the dataset is highly imbalanced. Without taking this into account, a model might perform well on non-default cases but fail to recognize actual defaulters.

To reduce this problem, different class-imbalance settings are used for the four models:

- **Logistic Regression:** Balanced class weights are applied.
- **Random Forest:** Balanced class weights are applied.
- **SVM:** Balanced class weights are applied.
- **XGBoost:** The scale_pos_weight parameter is used to give additional importance to the minority class.

3.3 Machine Learning Models Used

1. Logistic Regression
2. Random Forest
3. Support Vector Machine (SVM)
4. XGBoost

3.4 Evaluation Metrics

- **Accuracy:** Shows the percentage of all predictions that are correct.
- **Precision:** Tells us how many customers predicted as defaulters actually defaulted.
- **Recall:** Measures how many of the actual defaulters were correctly identified.
- **F1-Score:** Combines precision and recall into a single score.
- **ROC-AUC:** Measures how well a model distinguishes between default and non-default customers across different classification thresholds.

In this project, recall is particularly important because missing a customer who is likely to default may result in financial loss for the bank. ROC-AUC is also useful because it provides a broader view of how well the model separates the two classes.

4 Results

4.1 Performance Comparison

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	Logistic Regression	0.8550	0.1730	0.8806	0.2892	0.9487
1	XGBoost	0.8885	0.2023	0.7910	0.3222	0.9374
2	SVM	0.8410	0.1617	0.8955	0.2740	0.9152
3	Random Forest	0.9615	0.4405	0.5522	0.4901	0.9053

Table 1: Performance Comparison of the Four Classification Mode

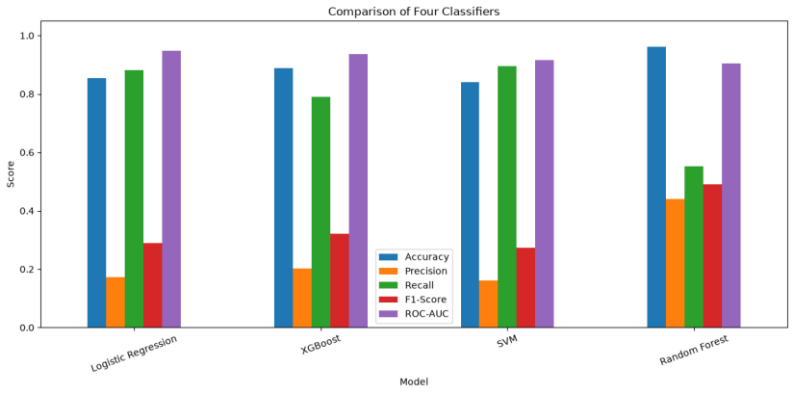


Figure 3: Side-by-side comparison of all five metrics across the four models.

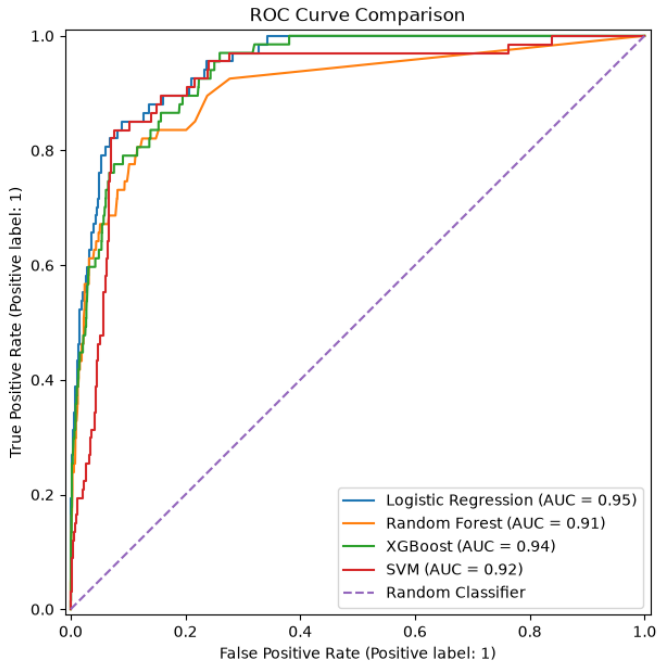


Figure 4: ROC curves for all four models. Logistic Regression has the largest area under the curve.

4.2 Selecting the Best Model

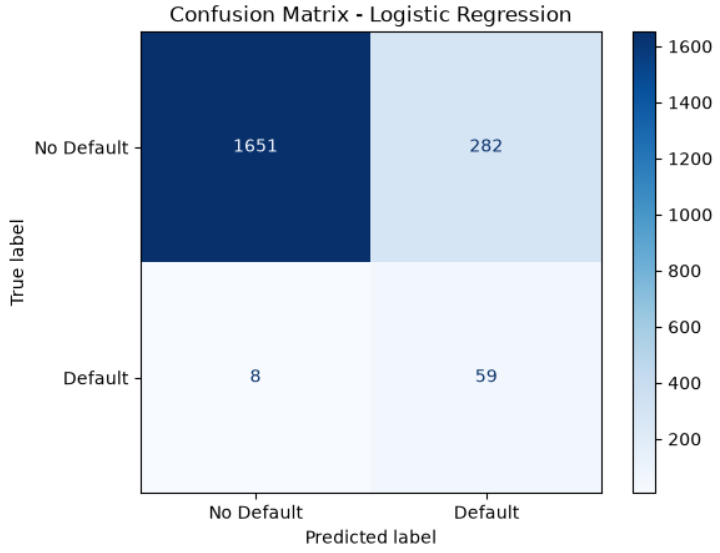


Figure 5: Confusion matrix for the selected best model, Logistic Regression.

5 Feature Importance

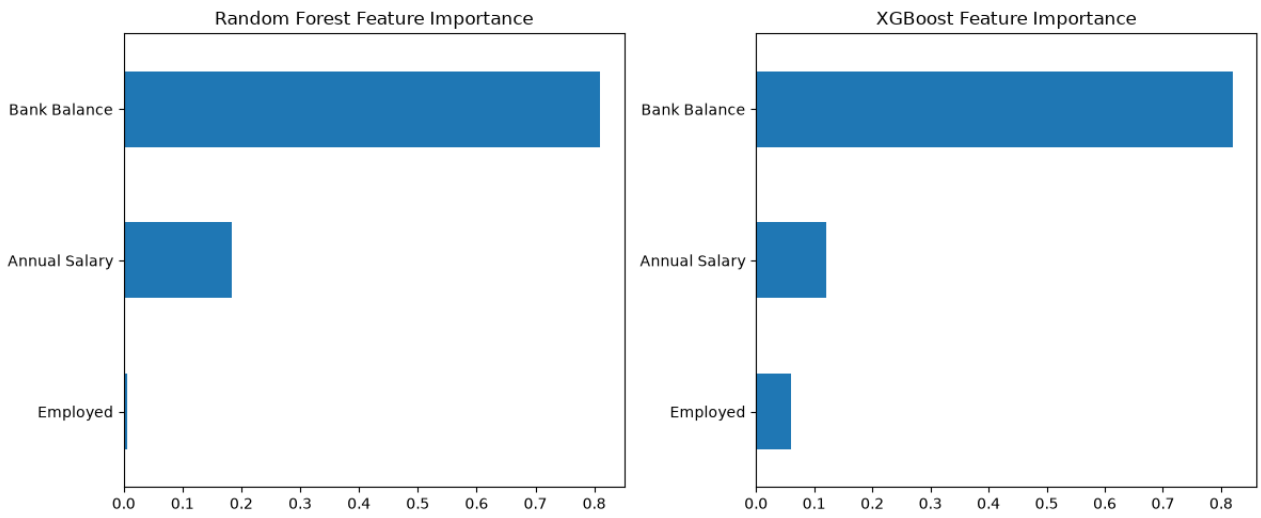


Figure 6: Feature importance from Random Forest (left) and XGBoost (right). Both models agree that bank balance dominates.

6 Conclusion

This project used four machine learning algorithms to predict loan default based on customer information. Among the models tested, Logistic Regression achieved the highest ROC-AUC score of 0.9487, while Random Forest recorded the highest accuracy of 96.15%.

Feature importance analysis showed that bank balance was the most influential feature, followed by annual salary. The project also demonstrated why accuracy alone is not enough when working with an imbalanced dataset.

Overall, Logistic Regression was selected as the best model for this experiment because of its strong ROC-AUC and recall. In the future, the system could be improved through further model tuning and testing on additional banking data.

Tools Used

- **Programming Language:** Python
- **Data Processing:** Pandas and NumPy
- **Data Visualization:** Matplotlib and Seaborn
- **Machine Learning:** Scikit-learn and XGBoost
- **Model Saving:** Joblib
- **Development Environment:** Jupyter Notebook in Visual Studio Code